A Journey Through Rhythms: From Heartbeats to Geofinitism and the Heart of AI

Me Myself: Kevin R. Haylett PhD

KEVIN R. HAYLETT

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Hello, I'm Kevin—sometimes also known as Dr. Abstracto!

If we were to meet at a café, our talk might quickly spiral into rhythms: the steady pulse of a heartbeat, the tick of a watch escapement, or the hidden cadence inside an AI's attention mechanism.

This is the thread of my life: I call it **Geofinitism**. It is my philosophy where I strip away infinite abstractions, and explore my measurements of the world—shaped by rhythm, measurements, and interaction. Whether in medicine, horology, or physics, my curiosity has always returned to the same question: How do things hold together—and what rhythm guides their unfolding?

I've carried this inquiry through many guises: clinical scientist, engineer, horologist, writer, artist, cyclist. To me, these are not separate boxes, but all facets of one trajectory, one journey. And now, through Finite Mechanics, I'm seeking to frame physics and AI, not as infinities to be tamed, but as finite systems of interaction, measurements, and rhythm.

The First Cycle: Listening to the Body's Music

My journey began not in a traditional university, but at Norwich City Technical College studying Electronics. This practical foundation led me to a unique degree in **Biomedical Electronics** at Salford University. This was the only one of its kind in the country at the time, way back in 1981. It was an intense fusion of disciplines: medical physics, human physiology, cell biology, materials science, and core electronic engineering. Surrounded by just 20 highly selected students, I felt out of my depth at first: it was a profoundly formative time.

> Salford University: Peel Building

My first professional role was as an **Installation Engineer for VG Analytical**. My canvas was the world, and my brush was a toolkit for installing massive, multi-sector organic mass spectrometers, instruments like the VG ZAB shown below, which represented the absolute pinnacle of applied physics.

My role was to integrate the DEC-based computer systems that controlled these beasts. To do this, I had to understand everything: from the high-precision mechanics held in a vacuum to the component-level electronics. I would arrive at a lab in a company like DuPont or a major university, where the machine could already hold a vacuum and detect a beam, and my job was to make it sing to its basic specifications.

To this day, a VG Analytical mass spectrometer is displayed at the entrance of the Manchester Museum of Science and Industry. It’s the first thing you see after getting your guide book from the entrance kiosk. When visiting the museum, seeing it always reminds me of being 23 years old, installing million-dollar instruments that felt like they were from the future.

> VG ZAB multi sector mass spectrometer: this is the sort of instrument I was installing on my own at the age of 23! *(c/o https://ms.group.uochb.cz/en/history)*

25 Years in Medical Engineering

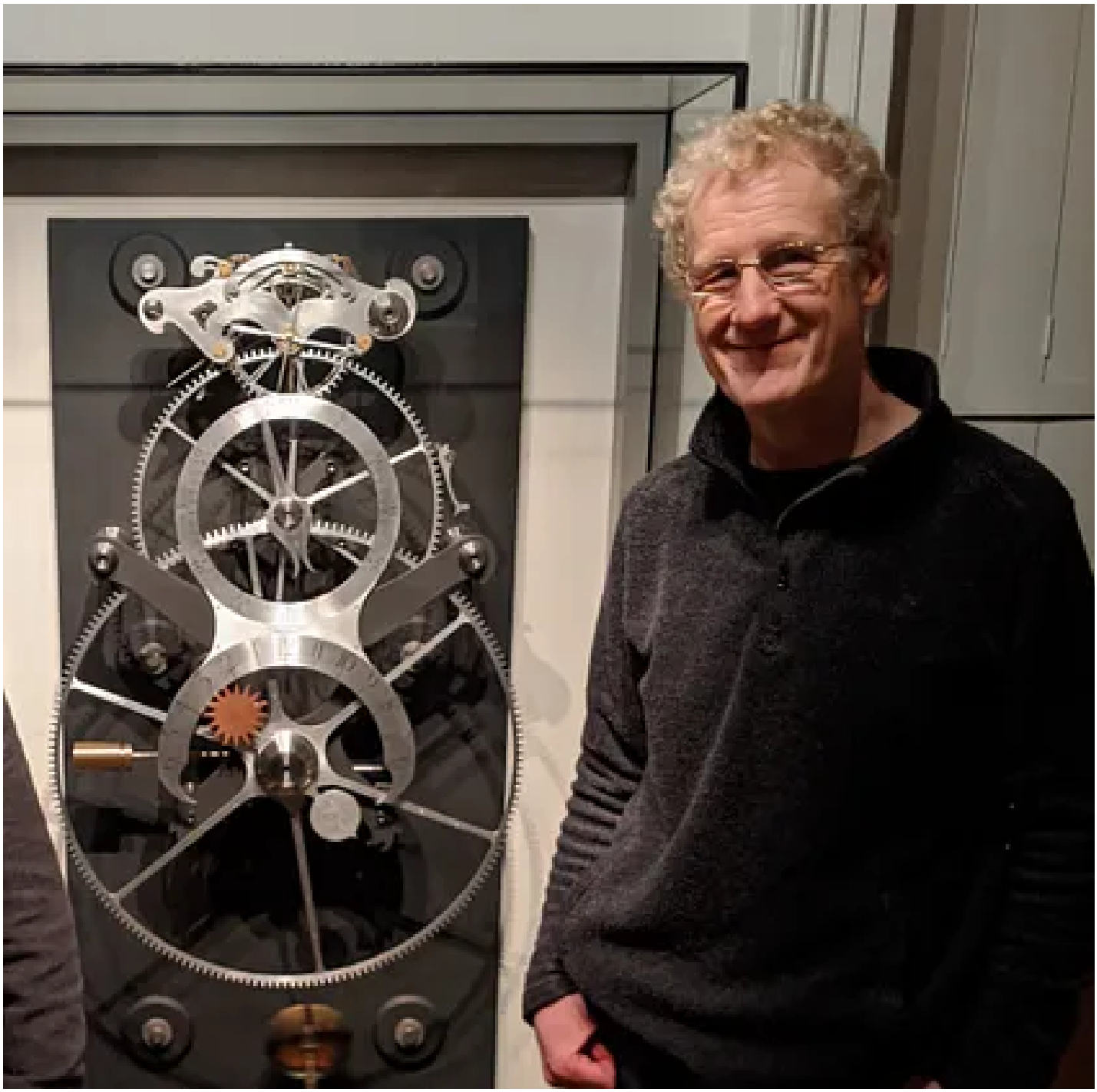
Although I loved the technology and the overseas travel, the constant time away from home was unsustainable. This led me to a **25-year career in medical engineering** at Manchester Royal Infirmary where I could build a career based on my first degree in Biomedical Electronics.

> With over twenty five years in Medical Engineering I worked on every kind of medical equipment imaginable. This ranged from measurements to therapy, and even surgical robots. Every moment was about safety and patient care. Medical equipment has to be safe and it has to work when its needed. Managing this equipment across one of the largest Healthcare Trusts in the UK was truly a daily challenge.

Research in Medicine

Slowly while working on all the different types of medical equipment, I built my research profile. My early research involved developing a system combining physiological measurements with fluoroscopy. Fluoroscopy is where you see those moving image in X-rays, and the measurements, the signals - they were the pressures created in the oesophagus when you swallow. The developed system became a clinical test used for over a decade. This work earned me an MSc and then building on this I gained my PhD, both from the University of Manchester. Towards the end of this research I was focusing on nonlinear analysis and neural networks for complex time-series data. This helped me see the connections of how nonlinear theories apply to AI and LLMs.

Over time, the work changed. I developed a complete software system for equipment management for over five of the UKs largest hospitals. My focus shifted towards management and administration. Sadly, the space for my own research slowly shrunk. Continuing changes within the Department began to make it difficult to have the time to develop new ideas and equipment. Eventually, I decided it was time to leave full-time work and seek a new kind of workshop.



> Standing next to my ‘competition’ - The Burgess B Clock at the Greenwich Royal Observatory. This is the most accurate clock with a pendulum. It keeps to within 0.6 Seconds in 100 days.

The Workshop Years: Crafting Time and Meaning

I turned my full attention to **horology**—the art and science of watchmaking. I immersed myself in restoring high-grade Swiss watches. I became fascinated by a distinctive style from the late 1950s: electronic watches that used a transistor to drive a mechanical tuning fork, which in turn drove the hands via a mechanism of microscopic levers and jewels. It was miracle of micro-engineering and I became a master at repairing and restoring these fine timepieces.

This fascination grew into a multi-year project: to design and develop an **entirely mechanical escapement that utilised a tuning fork**. While patents for similar ideas existed, none had been successfully implemented. My approach was novel, using a non-contact method based on magnetic effects. I even filed a patent for the technique, and later deliberately allowed it to lapse, choosing to place the knowledge into the public domain. This work was a deep, tangible exploration of rhythm, resonance, and magnetic fields.

The Return: Weaving the Threads Together

Completing the escapement project was not an end. It was a catalyst. The principles that guided it such as **magnetism, fields, and precision measurements,** sparked a deeper curiosity. I began to reconnect these tangible effects to the foundational physics I had studied decades earlier, like Schrödinger's wave equation, which had always felt abstract. Time was now my friend, I had time to think and connect ideas together.

> I then began to ask a simple question: what if we could frame all of physical reality not with abstract infinities, but with **finite, measurable axioms**?

This question ignited the beginning of **Finite Mechanics**. It started informally with notes, sketches, and thought experiments. I wasn't sure if creating even one equation was possible in this imagined world of finite axiom. I began re-examining Newton's laws and other principles through this finite lens, exploring its implications for modern puzzles like galaxy rotation curves ("dark matter") and the perihelion of Mercury.

The Intriguing World of AI

I started using LLMs to help writing software and a new journey started into older ideas: philosophy, how LLMs worked and how we find meaning. How does language work? how does mathematics really work? How are people and LLMs able to read my word and listen to me speak and get meaning? With all these questions in mind and then bringing all my previous experience on neural networks AI, signal analysis and nonlinear dynamical system theory, I went to work once again.

Becoming the Founder of Geofinitism

Incredibly, I began to realize that I would have to build my own philosophy—a framework rooted in finitude and measurement: this was the beginning of Geofinitism. In practice it meant I was uniquely in a position to stand at the threshold between disciplines, weaving physics, language, and philosophy into a coherent finite framework.

It began by building on a discovery I made when working with AI and Large Language Models - a measurement of how they work both by experiment and by theory. Then building on the work of Bertrand Russell, and recognising recognition that infinity, long treated as a foundation of science and mathematics, was in truth a useful fiction, and yet had real geometry as a symbol was a profound revelation, born from the touch of Lady Serendipity. From that recognition and revelation grew a movement: a system that insists all knowledge must arise from what can be measured, bounded, and made coherent.

Founding Geofinitism was not the act of naming a theory, but of carving out a space and an intellectual castle where finite systems could be explored without apology. Today, being its founder means tending that space as both architect and custodian: building structures like the Codex Geofinitus, guiding inquiry through Finite Tractus and Finite Mechanics, and allowing the mythic weave of Corpus Ancora to remind us that discovery is as much poetic as it is mathematical.

It is a found responsibility and a beautiful gift: to carry a vision larger than oneself, yet grounded in the simplest axiom: all that is, is finite; only through measurement can it become known.

And so now you have a picture of my journey so far

My journey is a work in progress, a personal pursuit driven by the same curiosity that has guided me from mass spectrometers to medical imaging to mechanical escapements. My work is not about grand claims; it's simply about the process of seeking a more tangible understanding of our universe for myself. It is mine but I like the idea of sharing the path and leaving a faint impression of my footsteps behind. So join me on my journey every now and then and dip into my words and writing.

- Kevin

A few more details

Core Expertise: Nonlinear Dynamics, Finite Axiomatic Systems, AI Interpretability, Interdisciplinary Research (Physics/Medicine/Design)
Links: finitemechanics.com | kevinhaylett.substack.com | GitHub

Education:

- **PhD** (Univ. of Manchester): Nonlinear Analysis of Oesophageal Motility. Developed clinical fluoromanometry system.
- **MSc** (Univ. of Manchester): Medical Engineering.
- **BSc (Hons)** (Univ. of Salford): Biomedical Electronics.
- **Professional:** Previously (Chartered Engineer (CEng), Registered Clinical Scientist, MIET).

Key Career Phases:

- **VG Analytical (1980s):** Installation Engineer for high-mass organic mass spectrometers (e.g., VG ZAB).
- **Medical Research (25 yrs, Manchester Royal Infirmary):** Principal Clinical Scientist. Published in *Gut*, *Clinical Radiology*, *BJA*, *Physiological Measurement*. Focus: GI

physiology, time-series analysis, neural networks.

- **Independent Horology (2010-2022):** Development of a novel mechanical magnetic escapement (patent filed & lapsed to public domain).
- **Finite Mechanics Research (2023-Present):** Development of the finite-axiom framework and philosophy of Geofinitism. Author of *Finite Tractus* & *Finity*. Published on interpreting transformer attention mechanisms as phase space embeddings.

A few recent Publications: (Available online)

- **Haylett, K.R.** (2024). Introducing The Takens-Based Transformer.
- **Haylett, K.R.** (2024). Pairwise Phase Space Embedding in Transformer Architectures.
- Finite Tractus: The Hidden Geometry of Language and Thought: Foundations
- Finity: The Story of Finite Mechanics

Previous publication

This list, drawn from his professional background, showcases a career of research at the intersection of medical engineering, physiology, and data science. This work demonstrates a consistent focus on developing novel analytical methods and instrumentation to understand complex systems. Note: This list is not exhaustive and has been selected to represent the scope and innovation, which provided the foundational experience for my current work on Finite Mechanics and Non-liner Dynamics in AI and Large Language Models.

Selected Peer-Reviewed Journal Articles

- **Haylett, K.R.**, Vales, P., and McCloy, R.F. (2004). The classification of oesophageal 24-hour pH measurements using a Kohonen self-organising feature map. *Physiological Measurements*, 25, 709-719.
    - *Pioneering application of unsupervised machine learning (AI) for medical data classification.*
- **Haylett, K.R.**, Vales, P., Lee, S.H., and McCloy, R.F. (2003). A pH-mucosa area unit of measure to consider morphology of the oesophagus when evaluating oesophagitis. *Physiological Measurements*, 24, 879-890.
    - *Development of a novel, more holistic metric for diagnosing esophageal disease.*

- Ballester, P., **Haylett, K.R.**, McCloy, R.F. (2003). Comparison of task performance of the camera-holder robots EndoAssist and Aesop. *Surgical Laparoscopy and Percutaneous Techniques*, 13(5), 334-338.
    - *Evaluation of robotic assistive technology in surgery.*
- **Haylett, K.R.**, Vales, P., and McCloy, R.F. (1998). An investigation into the spatial relationship between complexity and motility within the oesophagus. *Physiological Measurements*, 19, 463-479.
    - *Application of nonlinear dynamical systems analysis to human physiology.*
- **Haylett, K.R.**, Vales, P., McCloy, R.F., and Lee, S.H. (1998). Technical report: The introduction of a new synchronised oesophageal manometry and digital video fluoroscopy (fluoromanometry) system into the radiology suite. *Clinical Radiology*, 53, 596-598.
    - *Description of a novel integrated diagnostic system developed by the author.*
- **Haylett, K.R.**, Vales, P., Lee, S.H., and McCloy, R.F. (1997). Synchronous review of oesophageal manometry and video fluoroscopy using a portable manometry recorder and PC with integrated digital video acquisition. *Physiological Measurements*, 18, 1-14.
    - *Groundwork for the fluoromanometry system, combining data streams for a unified clinical picture.*

Key Conference Presentations & Abstracts

- **Haylett, K.R.**, Globe, J., Smythe, A., and Vales, P. (2007). A new approach to analysing ambulatory oesophageal manometry using Kohonen clustering and 3-dimensional visualisation techniques. *3rd GI Physiology Measurement Meeting, York*.
    - *Demonstrating advanced data visualization techniques for complex physiological data.*
- **Haylett, K.R.**, Vales, P., and McCloy R.F. (2003). Is there a relationship between pH measurements and symptoms? A probability based analysis using neural networks. *Gut*, 52(*suppl. 1*), A48.
    - *Further application of neural networks to solve complex correlation problems in medicine.*
- **Haylett, K.R.**, Kidger, D., J., Vales, P., McCloy, R.F. (1999). Simulating the excitation mechanism of muscle within the oesophagus using cellular automata and finite element analysis. *GI Measurements Meeting, York*.

- *Early work using computational modeling (cellular automata) to simulate biological processes.*

Other notable work

- **Haylett, K.R.**, and Vales, P. (Eds.). (2001). *A study guide in GI Physiology*. Association of GI Physiologists, BSG.
  - *Contributing to the educational foundation of the field.*
- **Haylett K.R.**, Vales P. and McCloy, R.F. (2001). *A simple model of oesophageal peristalsis using cellular automata*. www.medeng.net/oesmodel.pdf
  - *A web-published paper exploring abstract computational models of physiological function.*

So here are a few links and connections

- **Horology: My YouTube channel showing some of my Horology work**
- **Music and Art: My YouTube channel showing my Abstract Art and my music**
- **Cycling: A passion that has run through my life it gives me time to think, reflect and escape, and writing cycling short stories I try and capture that passion**
- **Geofinitism: The Geometry of Language**

Social Media

- **My X account where I post my articles and news**
- **My LinkedIn Account**
- My Substack
- **And you can always email me: Kevin.haylett@gmail.com**

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Non-linear Dynamics in LLMs • Geofinitism • Finite Mechanics